



भारतीय प्रौद्योगिकी संस्थान (भारतीय खनि विद्यापीठ)

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INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES)
DHANBAD - 826004, JHARKHAND, INDIA

मध्यावधि परीक्षा (Mid Semester Examination)

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Subject : Rock Engineering Part : 1 Semester : 6th

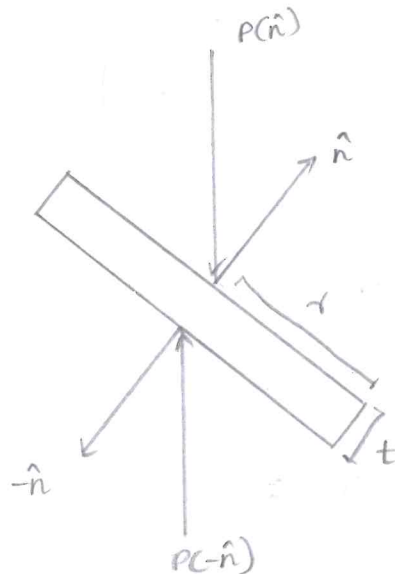
Signature of Invigilator : [Signature] 23/02/25 Date : 23/2/25

Question No.	1	2	3	4	5	6	7	8	9	10	TOTAL
Marks Obtained											

Total Marks Obtained : _____ Signature of Examiner : _____

Ques 1)
(a)

Cauchy's First law of stress →



t = thickness of the coin-shaped body of a continuum potato

r = Radius of the penny shaped object

force acting on the body on the face with the normal vector ($\hat{n}$) →
 $= p(\hat{n}) \times \pi r^2$

force acting on the body on the face with the normal vector ($-\hat{n}$) →
 $= p(-\hat{n}) \times \pi r^2$

force acting on the Rim
 $= \pi r^2 h \times t$

(t = cumulative force acting on the width of the coin)

If the body is in equilibrium →

$$p(\hat{n}) \times \pi r^2 + p(-\hat{n}) \times \pi r^2 + \pi r^2 h \times t = 0$$

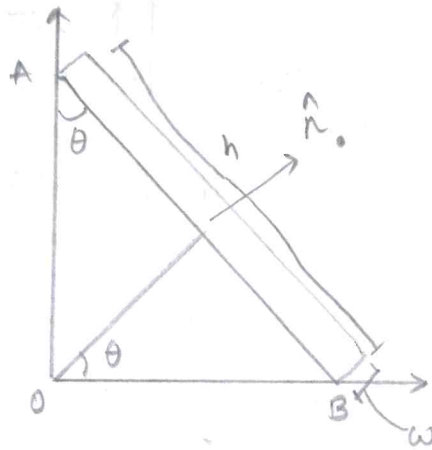
$$\text{as } h \rightarrow 0 \\ \pi r^2 h \rightarrow 0$$

$$p(\hat{n}) \times \pi r^2 + p(-\hat{n}) \times \pi r^2 = 0$$

$$\therefore p(\hat{n}) \times \pi r^2 = -p(-\hat{n}) \times \pi r^2 \quad \text{as } \pi r^2 \neq 0,$$

$$\therefore \boxed{p(\hat{n}) = -p(-\hat{n})}$$

ii)
(b) Proof for Cauchy's second law of stress $\rightarrow$



proof for $\tau_{xy} = \tau_{yx}$ Cauchy's second law of stress $\rightarrow$

consider a body with hypotenuse $= h$ and width $= w$,

finding force on the side AB

$$= p(\hat{n}) \times w \times h$$

finding force on the side OB

$$= p(-\hat{e}_y) \times w \times h \sin \theta$$

finding force on the side OA

$$= p(-\hat{e}_x) \times w \times h \cos \theta$$

(3)

as the body is overall in equilibrium state,

$$p(\hat{n}) \times w \times h + p(-\hat{e}_y) \times w \times h \sin \theta + p(-\hat{e}_x) \times w \times h \cos \theta = 0$$

$$\therefore p(\hat{n}) + p(-\hat{e}_y) \sin \theta + p(-\hat{e}_x) \cos \theta = 0$$

$$\text{as } [-p(\hat{n}) = p(-\hat{n})] \text{ from Cauchy's first law,}$$

$$p(\hat{n}) = p(\hat{e}_y) \sin \theta + p(\hat{e}_x) \cos \theta$$

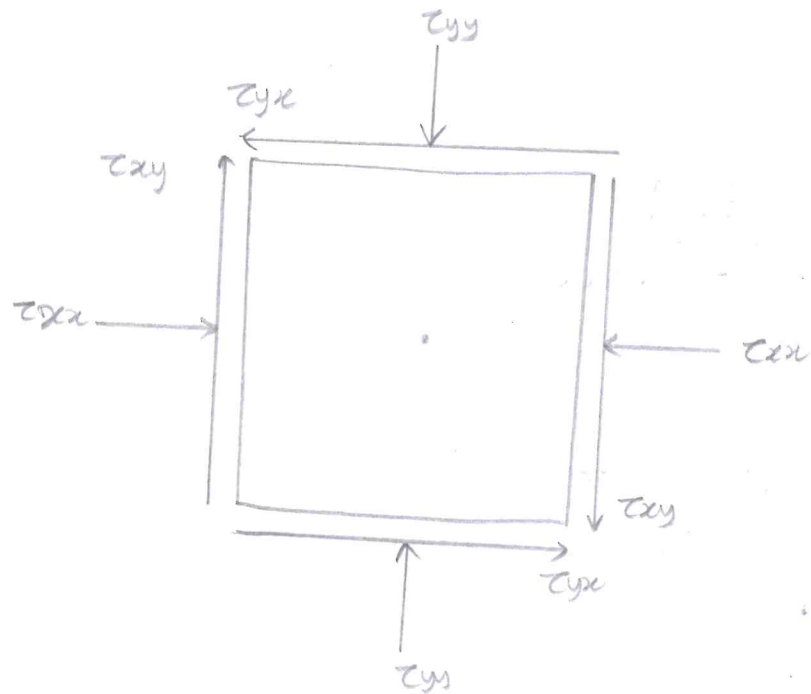
$$\text{as } n_x = \cos \theta$$

$$n_y = \sin \theta$$

$$\therefore p(\hat{n}) = p(\hat{e}_y) n_y + p(\hat{e}_x) n_x$$

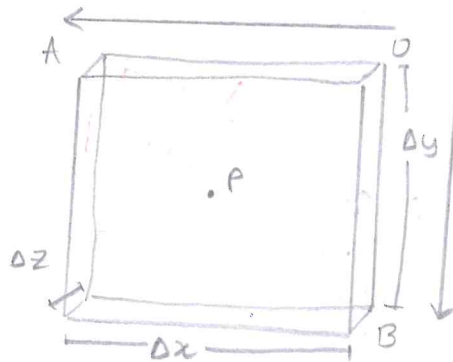
— hence proved.

Ques 1)
(b)



Proof for $\tau_{xy} = \tau_{yx}$,

Let us consider a small element of dimension Δx , Δy and Δz



Force acting on face AB $\rightarrow$

$\tau_{xx} \times \Delta y \times \Delta z =$ in normal direction

$\tau_{xy} \times \Delta y \times \Delta z =$ in shear direction

Momentum on face AB with respect to centre (P),

As τ_{xx} passes through the centre there is no momentum.

$\tau_{xy} \Delta y \times \Delta z \times \frac{\Delta x}{2} =$ Momentum in clockwise direction
Moment!

Force acting on face OA $\rightarrow$

$\tau_{yy} \times \Delta x \times \Delta z =$ in normal direction.

$\tau_{yx} \times \Delta x \times \Delta z$ in shear direction

Momentum on face OA with respect to centre (P) $\rightarrow$

As τ_{yy} passes through the centre its momentum is zero,

$\tau_{yx} \times \Delta x \times \Delta z \times \frac{\Delta y}{2} =$ momentum in anticlockwise direction.

Conserving momentum

$$\tau_{xy} \Delta x \Delta y \Delta z - \tau_{yx} \Delta x \Delta y \Delta z = 0$$

3

$$\therefore \tau_{yx} = \tau_{xy} \text{ as } \frac{\Delta x \times \Delta y \times \Delta z \neq 0}{2}$$

$$\therefore \boxed{\tau_{yx} = \tau_{xy}}$$

— hence proved.

Ques 1)

(C)

If normal plane makes,

$\angle 30^\circ$ with x axis and $\angle 45^\circ$ with y axis,

$$n_x = \frac{1}{2} \sqrt{3}$$

$$n_y = \frac{1}{\sqrt{2}}$$

$$n_z = ?$$

$$\text{as } n_x^2 + n_y^2 + n_z^2 = 1 \quad n_z = -1/4$$

$$\frac{3}{4} + \frac{1}{2} + n_z = 1$$

$$\frac{3}{4} + \frac{2}{4} + n_z = 1$$

$$\therefore n_z = -1/2$$

Question was wrong!

$$\text{traction} = \begin{bmatrix} 15 & 5 & 6 \\ 5 & 5 & 4 \\ 6 & 4 & 10 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 \\ 1/\sqrt{2} \\ -1/2 \end{bmatrix}$$

$$= \begin{bmatrix} 15\sqrt{3}/2 + 5/\sqrt{2} - 6/2 \\ 5\sqrt{3}/2 + 5/\sqrt{2} - 4/2 \\ 6\sqrt{3}/2 + 4/\sqrt{2} - 10/2 \end{bmatrix} = \begin{bmatrix} 13.525 \\ 5.865 \\ 3.024 \end{bmatrix}$$

$$\text{Normal stress} = p(\hat{n}) \cdot [\hat{n}]$$

$$= \begin{bmatrix} 15\sqrt{3}/2 + 5/\sqrt{2} - 6/2 \\ 5\sqrt{3}/2 + 5/\sqrt{2} - 4/2 \\ 6\sqrt{3}/2 + 4/\sqrt{2} - 10/2 \end{bmatrix} \cdot \begin{bmatrix} \sqrt{3}/2 \\ 1/\sqrt{2} \\ -1/2 \end{bmatrix}$$

$$= \frac{45}{4} + \frac{5\sqrt{3}}{2\sqrt{2}} - \frac{3\sqrt{3}}{2} + \frac{5\sqrt{3}}{2\sqrt{2}} + \frac{5}{2} - \frac{2}{\sqrt{2}} - \frac{6\sqrt{3}}{4} + \frac{4}{2\sqrt{2}}$$

$$= \left(\frac{45 - 6\sqrt{3} + 10}{4} \right) + \left(\frac{10\sqrt{3} + 5\sqrt{3} - 4}{2\sqrt{2}} \right) + \left(\frac{5 - 3\sqrt{3}}{2} \right) - \sqrt{2}$$

$$= 11.152 + 4.709 + (-0.098) - \sqrt{2}$$

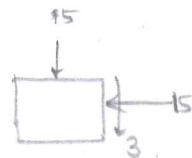
$$= 14.748$$

$$|\text{traction}| = |p| = 15.0488.$$

$$|p|^2 = |\tau|^2 + |\sigma|^2$$

∴ magnitude of normal stress = ~~14.348~~ ~~14.6887~~ 14.348

shear stress = ~~4.539~~ ~~3.272~~
4.539



Q403)

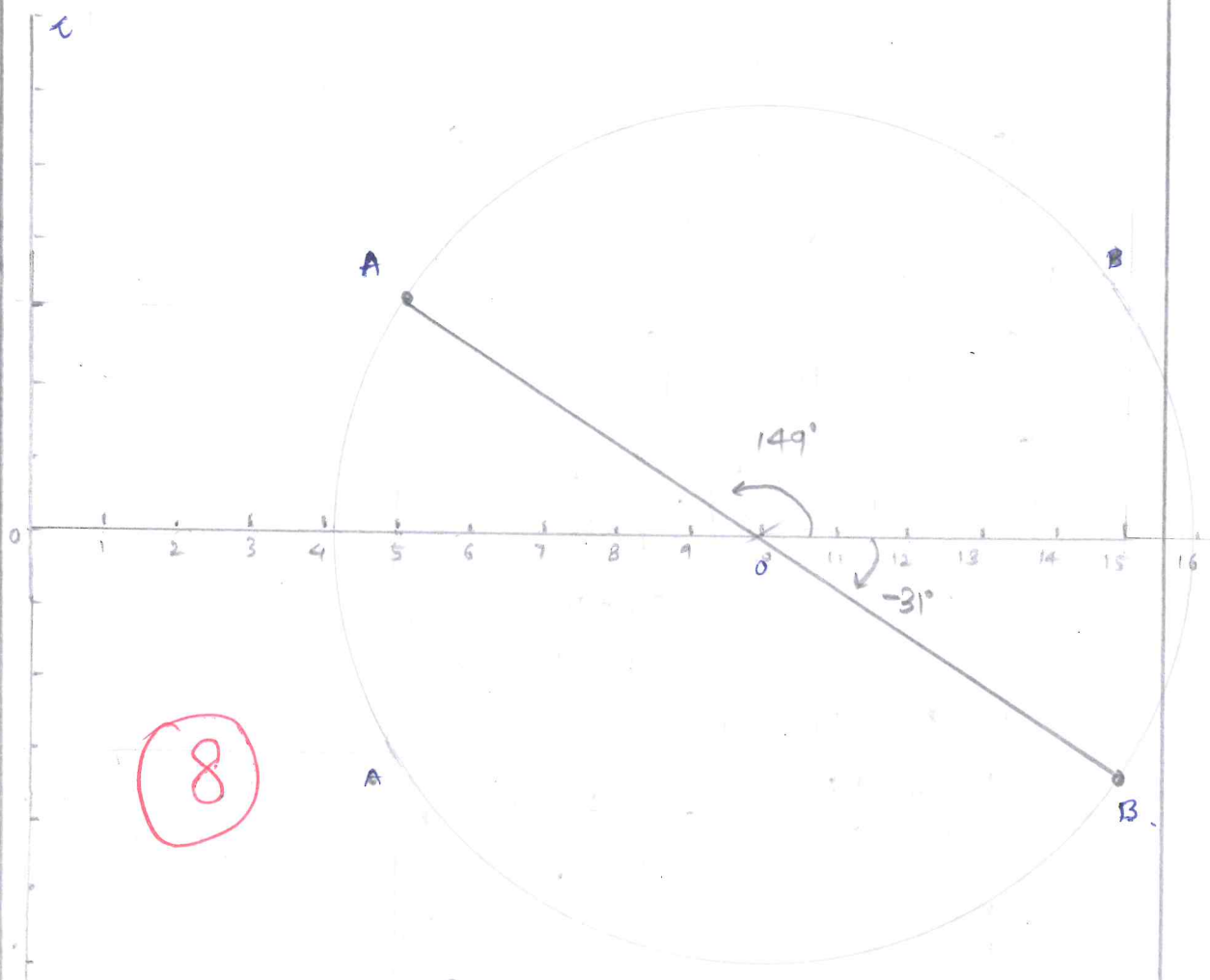


fig: mohr's circle,

$$\tan \theta = \frac{3}{55}$$

In mohr's circle,

$$\sigma = \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) + \frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau = \frac{1}{2}(\sigma_{xx} - \sigma_{yy}) \sin 2\theta + \tau_{xy} \cos 2\theta$$

So much of Calculation is not Required for Mohr's Circle

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_2) + \frac{1}{2}(\sigma_1 - \sigma_2) \cos 2\theta$$

$$\tau = \frac{1}{2}(\sigma_1 - \sigma_2) \sin 2\theta$$

The values could have been read from.

using protractor we see that $\theta = -31^\circ$ approximately, for σ_1 the

Mohr's Circle itself!

$$r \cos \theta = 5$$

$$\therefore r = 5.83 \quad a \approx 10$$

$$a = \frac{\sigma_1 + \sigma_2}{2}$$

$$\therefore \sigma_2 = a - r = 10 - 5.83 \sim 4.169$$

$$\sigma_1 = a + r = 10 + 5.83 \sim 15.830$$

for σ_2 $\theta \approx +149^\circ$

$$\begin{bmatrix} 15 & 3 \\ 3 & 5 \end{bmatrix}$$

$$|A - \sigma I| = 0$$

$$\begin{vmatrix} 15-\sigma & 3 \\ 3 & 5-\sigma \end{vmatrix} = 0$$

$$(15-\sigma)(5-\sigma) - 9 = 0 \quad 75 - 20\sigma + \sigma^2 - 9 = 0$$

$$\sigma_1 = 15.83$$

$$\sigma_2 = 4.169$$

$$\sigma^2 - 20\sigma + 66 = 0$$

①

for $\sigma_1 = 15.83$

0.2766

$$\begin{bmatrix} -0.83 & 3 \\ 3 & -10.83 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -0.83 & 3 & | & 0 \\ 3 & -10.83 & | & 0 \end{bmatrix} = \begin{bmatrix} -0.83 & 3 & | & 0 \\ 0.83 & -3 & | & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -0.83 & 3 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$-0.83 n_x + 3 n_y = 0$$

$$\text{If } n_y = 1k$$

$$n_x = 3.614k$$

$$\begin{bmatrix} 1 \\ 3.614 \end{bmatrix}$$

Rank=1

$$= \begin{bmatrix} 0.2667 \\ 0.9637 \end{bmatrix}$$

③

Normalising

$$n_x = \cos \theta$$

$$n_y = \sin \theta$$

$$\tan \theta = \left(\frac{1}{3.614} \right) \quad \theta = 15.466^\circ$$

$$\boxed{n_1 = 0.2667 \hat{i} + 0.9637 \hat{j}}$$

for $\sigma_2 = 4.169$

$$\begin{bmatrix} 10.831 & 3 \\ 3 & 0.831 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 10.831 & 3 & | & 0 \\ 3 & 0.831 & | & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 10.831 & 3 & | & 0 \\ 10.831 & 3 & | & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Rank=1

$$\begin{bmatrix} 10.831 & 3 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\therefore 10.831 n_x + 3 n_y = 0$$

$$\text{If } n_x = 1k$$

$$n_y = -3.614k \quad \begin{bmatrix} 1 \\ -3.614 \end{bmatrix}$$

$$n_x = \cos \theta$$

$$n_y = \sin \theta$$

$$\tan \theta = \left(\frac{-3.614}{1} \right) = -74.533^\circ \text{ in CW}$$

$$= 105.466^\circ \text{ in ACW}$$

$$\text{normalising} = \begin{bmatrix} 0.2667 \\ -0.9637 \end{bmatrix} = n_2 = \boxed{0.2667 \hat{i} - 0.9637 \hat{j}}$$

③

Comparing Mohr's circle and calculation

Mohr's circle

$$\sigma_1 = 15.83$$

$$\theta_1 = -31.1^\circ$$

$$\sigma_2 = 4.169$$

$$\theta_2 = 19.9/2^\circ$$

calculation

$$\sigma_1 = 15.83$$

$$\sigma_2 = 4.169$$

$$\theta_1 = 15.466^\circ$$

$$\theta_2 = -74.533^\circ$$

$$n_1 = 0.2667i + 0.9637j$$

$$n_2 = 0.2667i - 0.9637j$$

Correspondance of Mohr's θ to Normal θ .

Mohr's circle $2\theta \rightarrow \theta$

ACW must be made CW

and hence they are very similar values

Ques 2)

- (a) The Coal Brook disaster was a major accident that occurred in the Coal Brook coal mines that caused the initiation of a intensive Research Programme resulting in major advances in methods used in designing coal pillars

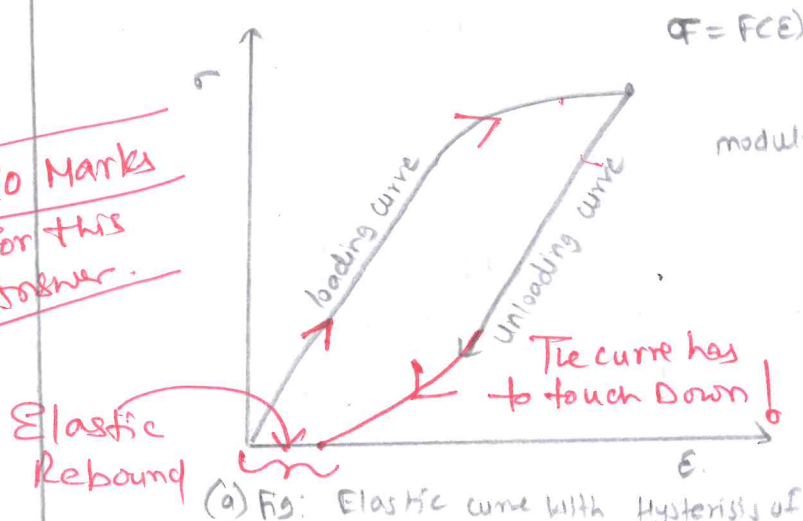
(b) Pillar strength = $\frac{K \times H^2}{w \times H^2}$
 $= K \times \frac{w}{H^2}$

H = height of pillar
 w = width of pillar
 $\alpha, \beta, k = \text{constants}$

- (c) If the pillar height is increased then the pillar strength Reduces. keeping width constant

(cd)

No Marks For this answer.



(a) Fig: Elastic curve with Hysteresis of Rocks

modulus = $\frac{d\sigma}{d\epsilon}$ The permanent deformation seen in the rock after unloading is known. Energy of loading > the Energy of unloading. Elastic Rebound

- (ce). We see that In rocks they tend to behave elastically with hysteresis which causes difference in loading and unloading curves and this phenomenon is called the elastic rebound

(f) This hysteresis caused is due to

Energy Required for Rock Loading > Energy Required for Rock Unloading.

this is because during loading some amount of energy is lost in cracking of the rocks or expansion of the cracks which is not regained during the unloading of the rock causing energy difference. ~~The Compaction of the existing cracks results in elastic rebound. They do not regain the original configuration when load is removed~~

g) Difference Between "conventional Engineering Design" and "Rock Engineering Design"

Conventional Engg. Design	Rock Engineering Design
(i) The first task is the load to assess the load being applied on the structure. (ii) Then accordingly decide the Required CHILE material to be used. C - continuous H - Homogeneous I - Isotropic L - Linearly elastic E ✓ (iii) Then decide upon the geometry of the structure to be designed (iv) It focuses on Enhancing yield.	(i) first examine the type of material that is being dealt with (usually) → D - discontinuous I - Inhomogeneous A - Anisotropic N - Nonlinearly elastic. E ✓ (ii) Then We know that there is no choice of material being used. (iii) There is no need of understanding load because its gets distributed uniformly (A) (iv) Understanding the failure of rocks and its stability is the focus (v) The geometry of structure / excavation depends upon the geology of surrounding area

Externally applied loads are not as important as the re-distributed load due to excavation!